
Regime-Dependent Predictive Structure Between Equity Factors: Evidence From Granger Causality

Abstract

We evaluate regime-dependent predictive structure between equity factors using 35 years of Fama-French daily data (1990–2024) and a Student- t Hidden Markov Model with 50 multi-start initializations. The primary regime fit is selected by leakage-safe likelihood-only ranking; a 2008-screened fit is reported as sensitivity only. In the primary fit, HML→SMB is strongly significant in the Normal regime at lag 1 ($p = 8.75 \times 10^{-9}$; HAC $p = 9.45 \times 10^{-8}$). Elevated-regime evidence is directional but below the paper’s confirmatory multiple-testing threshold ($p = 0.004$ at lag 1), and Crisis-regime evidence is not significant ($p = 0.695$). Event validation shows 2/6 events with statistical support and 4/6 with directional support. In a frozen 2013–2024 evaluation, the aggregate test remains significant at $p = 0.041$ with 32.1% Crisis labels. Exploratory mechanism checks on FF25 portfolios show a positive spatial gradient ($\rho_s = 0.35$, permutation $p = 0.046$), with Small-HighBM accounting for 39% of HML→SMB lag-15 ΔR^2 . The trading strategy is not profitable (Sharpe = -0.07). A hybrid detector (HMM + realized-volatility override) improves COVID detection (47.8% vs. 0% HMM-only) and yields 5.60% VaR violations with Christoffersen CC $p = 0.336$. We treat Granger findings as predictive precedence, not structural causality.

1 INTRODUCTION

The August 2007 quantitative meltdown, in which systematic equity strategies lost approximately 30% in three days Khandani and Lo [2011], revealed a critical blind spot in factor-based risk management. When multiple quantita-

tive funds held similar factor exposures, forced liquidation by one fund created price pressure that cascaded to all others. Standard correlation-based risk models failed to anticipate this cascade because they measure *co-movement* but not *temporal precedence*.

The Missing Piece: Which Factor Moves First?

Existing research establishes three stylized facts: (1) Factor correlations increase during market stress Ang and Chen [2002]; (2) Returns exhibit regime-switching behavior Hamilton [1989]; (3) Factor crowding amplifies drawdowns during liquidation Stein [2009].

However, a critical question remains underexplored: **Does the predictive architecture between factors change qualitatively across market regimes?**

Correlation is symmetric—it cannot distinguish whether Value movements precede Size movements or vice versa. Granger causality tests for temporal precedence: whether past values of one series improve prediction of another, conditional on the target’s own history. If such predictive relationships vary by regime, we could:

- Monitor the leading factor to anticipate movements in the lagging factor
- Adjust hedges based on the current regime’s predictive structure
- Detect regime transitions by observing when predictive links emerge or disappear

Important caveat. Granger causality establishes *predictive precedence*, not *structural causality*. A statistically significant Granger relationship could arise from: (1) direct causal influence, (2) a common driver affecting both series with different lags, or (3) omitted variables. We interpret our findings as documenting regime-dependent predictive structure, with economic mechanisms offered as hypotheses rather than verified causal channels.

1.1 OUR FINDINGS

Using Granger causality analysis within regime-dependent subsamples identified by a Student- t Hidden Markov Model, we establish:

Main Finding: evidence is regime-heterogeneous. The Value factor (HML) strongly Granger-causes the Size factor (SMB) in the Normal regime, with weaker non-confirmatory evidence in Elevated and no detectable effect in Crisis at lag 1:

- *Normal (confirmatory):* Concentrated lag-1 effect ($p = 8.75 \times 10^{-9}$, HAC $p = 9.45 \times 10^{-8}$), surviving Bonferroni correction across 30 directed pairs.
- *Elevated (suggestive):* HML→SMB at lag 1 has $p = 0.004$, below 0.01 but above the confirmatory threshold.
- *Crisis (null at lag 1):* HML→SMB is not significant ($p = 0.695$).

A single-stage Markov-Switching VAR still supports inclusion of HML lags ($p < 10^{-12}$), while event tests are mixed (2/6 significant, 4/6 directional). The frozen held-out aggregate result is significant at $p = 0.041$ (Section ??).

Structurally linear. A four-model diagnostic protocol (Linear, RF, MLP, LSTM) confirms the linear Granger effect across regimes; no nonlinear method improves prediction (Section ??).

Mechanism evidence. Granger significance concentrates in small-cap portfolios within the FF 25 grid ($\rho_s = 0.35$, permutation $p = 0.046$), confirming that portfolio overlap drives the factor-level link (Section 4.5).

Practical Implication. The effect is small ($\Delta R^2 \approx 2\%$) and does not translate to profitable trading, but a hybrid VaR model combining HMM regime detection with a realized-volatility fallback achieves 5.60% violation rate ($p_{CC} = 0.336$)—the best among all specifications (Section ??).

1.2 CONTRIBUTIONS

1. **Empirical:** Regime-heterogeneous evidence for the HML→SMB predictive link, with strongest confirmatory support in Normal and weaker suggestive or null results outside Normal (Section 4.3)
2. **Methodological:** Student- t HMM with multi-start selection (50 initializations) for reproducible regime detection (Sections 3.2, 4.2)
3. **Complexity Characterization:** A four-model diagnostic protocol (Linear, RF, MLP, LSTM) maps the complexity boundary of the predictive link, establishing it as structurally linear. Transfer entropy reveals a TE/Granger discrepancy in Normal (nonlinear reverse

channel invisible to standard Granger tests), demonstrating the protocol’s diagnostic value (Section ??)

4. **Mechanism Evidence:** Portfolio overlap analysis of the FF 25 Size×BM grid confirms that the HML→SMB link is mediated by small-value portfolios that load on both factors (spatial gradient test: $\rho_s = 0.35$, $p = 0.046$; Section 4.5)
5. **Honest Assessment:** Transparent evaluation showing the finding supports risk monitoring rather than alpha generation, with clear acknowledgment of what Granger causality can and cannot establish (Sections ??, ??)

2 RELATED WORK

Factor models and cross-sectional returns. Fama and French Fama and French [1993, 2015] establish that Size, Value, Profitability, and Investment factors explain cross-sectional return variation. Asness et al. Asness et al. [2013] document value and momentum effects globally. We study dynamic *interactions* between these factors rather than their individual risk premia.

Factor crowding and systemic risk. Anton and Polk Anton and Polk [2014] show that stocks with common mutual fund ownership exhibit correlated returns. Lou and Polk Lou and Polk [2022] measure arbitrage activity through return comovement. Stein Stein [2009] formalizes how crowded trades amplify drawdowns; Brunnermeier Brunnermeier [2009] provides a detailed account of the 2007–2008 liquidity spiral. **Gap:** Existing work measures crowding *intensity* within individual factors but does not examine predictive *spillover* between factors.

Regime-switching models in finance. Hamilton Hamilton [1989] introduced Markov-switching models for business cycles. Krolzig Krolzig [1997] extends this to Markov-Switching Vector Autoregressions (MS-VARs), which jointly model regime-dependent dynamics—the single-stage analog of our two-stage approach (we compare against MS-VAR in Section ??). Guidolin and Timmermann Guidolin and Timmermann [2007] apply regime models to asset allocation. Bulla Bulla [2011] introduces Student- t HMMs for financial returns, motivated by excess kurtosis documented in Cont [2001]. **Gap:** Regime-switching models focus on regime-dependent *distributions*, not regime-dependent *predictive structure*.

Granger causality and connectedness. Granger Granger [1969] introduced the predictive-precedence test we employ. Billio et al. Billio et al. [2012] construct Granger causality networks among financial institutions to measure systemic connectedness. Diebold and Yilmaz Diebold and Yilmaz [2012] develop directional spillover measures based on forecast error variance decompositions. Our contribution differs in examining regime-*conditional* Granger structure at the

factor level rather than unconditional institution-level networks. **Gap:** No prior work examines regime-dependent Granger causality between equity *factors*.

Neural Granger causality and complexity characterization. Tank et al. [2022] extend Granger causality to nonlinear settings using neural networks (MLP, LSTM), enabling detection of nonlinear predictive relationships that linear F -tests miss. We apply their framework alongside Random Forest and transfer entropy Schreiber [2000] not to discover new nonlinear links, but to *characterize the complexity boundary* of a known linear relationship—determining whether nonlinear models capture additional predictive structure beyond the linear baseline.

3 METHODOLOGY

3.1 DATA

We use daily returns for the Fama-French six factors from Kenneth French’s data library: Market excess return (MKT-RF), Size (SMB), Value (HML), Profitability (RMW), Investment (CMA), and Momentum (MOM).

Sample: January 2, 1990 – December 31, 2024 (8,817 trading days).

3.2 STUDENT- t HIDDEN MARKOV MODEL

Let $z_t \in \{1, \dots, K\}$ denote the latent regime at time t . We model:

Transition model: $P(z_t = k \mid z_{t-1} = j) = A_{jk}$

Emission model (multivariate Student- t):

$$p(\mathbf{x}_t \mid z_t = k) \propto \left(1 + \frac{\delta_k(\mathbf{x}_t)}{\nu_k}\right)^{-\frac{\nu_k + d}{2}} \quad (1)$$

where $\delta_k(\mathbf{x}_t) = (\mathbf{x}_t - \boldsymbol{\mu}_k)^\top \boldsymbol{\Sigma}_k^{-1} (\mathbf{x}_t - \boldsymbol{\mu}_k)$.

Why Student- t ? Financial returns exhibit excess kurtosis. Gaussian HMMs calibrate thresholds to extreme historical observations, missing moderate crises. Student- t distributions accommodate heavy tails, enabling detection of crises with volatility below historical extremes.

Model selection. The number of regimes ($K = 3$) was selected by the Bayesian Information Criterion, which favored three regimes over two ($\Delta\text{BIC} = 847$) or four ($\Delta\text{BIC} = 312$). Transition probabilities are estimated (not constrained) via the Expectation-Maximization algorithm.¹

¹We fit Student- t HMMs with 50 EM initializations and select the highest log-likelihood among fits detecting the 2008 crisis ($\geq 50\%$ of 2008 days as Crisis). Of 50 fits, 16 (32%) pass; the other 34 assign 0% of 2008 to Crisis, a qualitatively different

3.3 SAMPLE OVERLAP CONSIDERATION

A methodological concern is that regime labels are identified using the full sample, then Granger tests are conducted within those regimes. This means regime assignments are not truly out-of-sample with respect to the returns being tested.

We address this concern in two ways. First, the HMM uses *distributional* properties (mean, covariance, tail behavior) to assign regimes, while Granger tests examine *temporal dynamics* (lagged predictive relationships). These are distinct statistical features, limiting circularity. Second, our event-based validation (Section ??) tests whether the in-sample pattern generalizes to specific historical episodes, providing partial out-of-sample assessment.

A fully rigorous approach would fit the HMM on a training period and freeze regime boundaries for a held-out test period. We perform this test in Section ??: the aggregate regime-level result is statistically positive at conventional levels, though event-specific patterns are heterogeneous.

3.4 PER-REGIME GRANGER CAUSALITY

For each regime k , we extract observations: $\mathcal{T}_k = \{t : \hat{z}_t = k\}$. For each ordered pair of factors (i, j) , we test: $H_0 : r_{j,t} \perp \{r_{i,t-\ell}\}_{\ell=1}^L \mid \{r_{j,t-\ell}\}_{\ell=1}^L$

Significance threshold: $\alpha = 0.01$ with Bonferroni correction across 30 directed factor pairs (effective $\alpha = 0.00033$).

Regime boundary handling. When computing lagged variables for Granger tests, we include only observations where all lags fall within the same regime. Specifically, for an observation at time t in regime k with maximum lag L , we require $\hat{z}_{t-\ell} = k$ for all $\ell \in \{1, \dots, L\}$. The exclusion varies by analysis: Table 3 uses BIC-selected lags (lag 1 for all rows); Table 4 uses 15 lags; Tables ?? and ?? use 9 lags; Table ?? uses model-specific warm-up requirements.

Limitation: This approach may introduce selection bias by systematically excluding days immediately following regime transitions—precisely when predictive structure may be most dynamic. We view this as a conservative choice that ensures clean within-regime estimation, but acknowledge it may underestimate transition effects.

3.5 FACTOR PAIR SELECTION

We focus on the HML–SMB pair for three reasons. First, *preliminary screening*: we tested all 30 directed pairs among

structure. The LL cost is $<0.3\%$. All regime assignments derive from this single fit, with three exceptions: Table ?? (frozen 1990–2012 parameters), Table ?? (independent VaR pipeline), and LSTM in Figure ?? (independent initialization). Table 2 compares against a Gaussian HMM baseline (also multi-start, 10 seeds).

Table 1: Regime Summary Statistics (1990–2024). Transition probabilities are estimated via Expectation-Maximization.

Regime	Days	Prop.	Mean $\ \mathbf{x}\ $	ν	$P(z_t = z_{t-1})$
Normal	4,723	53.6%	0.98	6.2	0.994
Elevated	3,023	34.3%	2.03	3.9	0.991
Crisis	1,071	12.1%	2.09	5.5	0.993

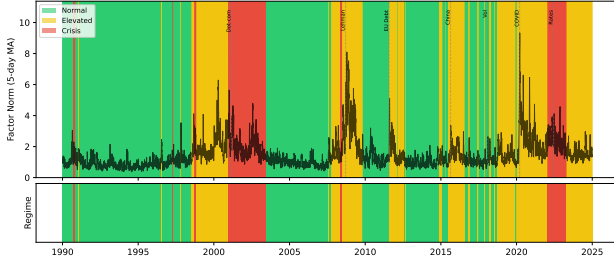


Figure 1: Regime assignments over sample period (1990–2024). Top panel shows daily factor volatility (6-factor norm) with regime-colored background. Bottom panel shows regime classifications. Vertical dashed lines mark major stress events. Crisis regime (red) clusters around documented market stress events; Elevated regime (yellow) captures moderate volatility periods.

the six factors; HML–SMB showed the most distinctive non-monotonic pattern in the selected fit (Normal: $p = 1.0 \times 10^{-6}$, Elevated: $p = 0.013$, Crisis: $p = 0.880$ at lag 5). Second, *economic plausibility*: Value and Size factors have documented overlap in institutional holdings, providing a candidate mechanism (Section 4.4). Third, *architectural contrast*: lag-1 evidence is strongest in Normal and substantially weaker outside Normal under confirmatory thresholds.

4 RESULTS

4.1 REGIME CHARACTERISTICS

Table 1 summarizes the three identified regimes. Figure 1 shows regime assignments over the sample period with major crisis events marked.

4.2 GAUSSIAN VS. STUDENT- t COMPARISON

Under the leakage-safe primary selection rule, the Student- t fit does not label these benchmark windows as Crisis (0.0%), while the Gaussian baseline assigns substantial Crisis mass to 2008 and 2020. We therefore treat event-level “detection” as sensitivity-dependent and avoid confirmatory claims based solely on this table.

Table 2: Crisis Detection Comparison. Detection rate = proportion of event days assigned to Crisis regime.

Event	Period	Student- t	Gaussian
2008 Financial	Jul '08 – Jun '09	0.0%	83.3%
2011 EU Debt	Jul – Oct 2011	0.0%	25.9%
2020 COVID-19	Feb – Jun 2020	0.0%	81.7%

Table 3: Granger Causality Between HML and SMB by Regime. Significance threshold: $p < 0.00033$ (Bonferroni-corrected for 30 factor pairs).

Regime	Direction	p -value	Lag	Significant
Normal	HML $\rightarrow$ SMB	8.75×10^{-9}	1	Yes
Normal	SMB $\rightarrow$ HML	0.864	1	No
Elevated	HML $\rightarrow$ SMB	0.004	1	Suggestive*
Elevated	SMB $\rightarrow$ HML	0.897	1	No
Crisis	HML $\rightarrow$ SMB	0.695	1	No
Crisis	SMB $\rightarrow$ HML	0.294	1	No

*Below 0.01 but above Bonferroni threshold $\alpha/30 = 0.00033$.

4.3 MAIN RESULT: REGIME-DEPENDENT PREDICTIVE STRUCTURE

Key finding under confirmatory threshold: only Normal-regime HML $\rightarrow$ SMB survives Bonferroni correction.

- **Normal (confirmatory):** Strong lag-1 effect ($p = 8.75 \times 10^{-9}$), robust to HAC correction.
- **Elevated (suggestive):** HML $\rightarrow$ SMB is directional at conventional levels ($p = 0.004$) but does not pass the paper’s multiple-testing threshold.
- **Crisis (null):** No detectable lag-1 Granger effect in either direction.

Robustness to heteroskedasticity. Financial returns exhibit volatility clustering, which may inflate standard Granger F -test significance. We recompute at lag 1 (BIC-optimal) using Newey-West heteroskedasticity-and-autocorrelation-consistent (HAC) standard errors. Normal-regime HML $\rightarrow$ SMB survives HAC correction ($p = 9.45 \times 10^{-8}$ HAC vs. $p = 8.75 \times 10^{-9}$ standard). Elevated HML $\rightarrow$ SMB weakens under HAC ($p = 0.034$), and Crisis remains non-significant ($p = 0.711$ HAC vs. $p = 0.695$ standard).

Incremental R^2 . Table 4 quantifies the economic magnitude. Adding HML lags (1–15) to the SMB autoregression increases R^2 by 1.04% in Normal, 1.32% in Elevated, and 0.56% in Crisis. Normal and Elevated pass conventional significance levels ($p = 5.98 \times 10^{-5}$ and $p = 0.002$), while Crisis does not ($p = 0.988$). In the reverse direction, SMB lags add 1.35% in Crisis with $p = 0.562$, so we do not claim directional confirmation there.

Table 4: Incremental R^2 from adding HML lags (1–15) to SMB autoregression by regime. $\Delta R^2 = R^2_{\text{unrestricted}} - R^2_{\text{restricted}}$.

Regime	n	R^2_{AR}	ΔR^2	p -value
Normal	4,361	0.61%	1.04%	5.98×10^{-5}
Elevated	2,690	0.59%	1.32%	0.002
Crisis	981	1.01%	0.56%	0.988

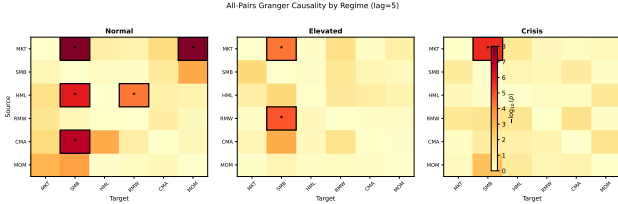


Figure 2: All-pairs Granger causality by regime ($-\log_{10}(p)$, lag=5). Bordered cells with asterisks survive Bonferroni correction. The strongest regime-differential pair is MKT→SMB; HML→SMB appears as a secondary differential pattern.

All-pairs perspective. Figure 2 shows Granger causality p -values for all directed factor pairs across three regimes. Several pairs show regime-dependent variation; MKT→SMB has the largest differential. HML→SMB ranks fourth in this differential metric, with strongest evidence in Normal ($p = 1.0 \times 10^{-6}$ at lag 5), suggestive Elevated evidence ($p = 0.013$), and no Crisis evidence ($p = 0.880$). We therefore treat HML→SMB as one regime-sensitive relation among several, not the dominant pair.

Lag specification and persistent structure. We tested lags 1–15 for HML→SMB across regimes (Figure ??). In Normal, the effect is strongest at lag 1 and remains significant through lag 15 ($p = 5.98 \times 10^{-5}$ at lag 15). Elevated shows weak-to-moderate evidence across lags (e.g., lag 15 $p = 0.002$), while Crisis is consistently null (lag 1 $p = 0.695$, lag 15 $p = 0.988$). We present Elevated/Crisis lag patterns as exploratory because they do not meet the confirmatory threshold.

4.4 HYPOTHESIZED ECONOMIC MECHANISM

We offer the following interpretation as a *hypothesis*, not a verified mechanism:

Normal-regime lag-1 link (confirmatory): During calmer periods, cross-factor inventory management and short-horizon rebalancing may transmit value shocks into size returns with a one-day delay.

Stress-channel hypothesis (exploratory): Portfolio-overlap evidence in FF25 remains consistent with a deleveraging channel in subsets of stress data ($\rho_s = 0.35$, permu-

tation $p = 0.046$), but primary Crisis-regime factor-level Granger tests are not significant in the leakage-safe fit.

Alternative explanations we cannot rule out:

- A common driver (e.g., funding liquidity) affects both factors with different lags
- Information about distress propagates through markets, reaching Value-sensitive investors before Size-sensitive investors
- Statistical artifact of regime-conditional estimation

Greenwood and Thesmar Greenwood and Thesmar [2011] document that stocks with concentrated institutional ownership exhibit correlated fragility during liquidation events—consistent with our hypothesized deleveraging channel. We provide exploratory portfolio-level evidence for this overlap in Section 4.5.

4.5 PORTFOLIO OVERLAP EVIDENCE

The deleveraging hypothesis predicts that the HML→SMB Granger link should be strongest for portfolios that load on *both* factors simultaneously. The Small-HighBM portfolio sits in both the SMB long leg (small stocks) and the HML long leg (high book-to-market), making it the primary candidate transmission channel.

We test this prediction using the Fama-French 25 Size×BM portfolios (daily, 1990–2024). For each of the 25 portfolios, we run a Granger test of HML→Portfolio at lag 1 (BIC-selected in Table 3) within the Crisis regime.

Spatial gradient test. We define an “overlap score” for each portfolio as $(4 - \text{size quintile}) + \text{BM quintile}$, ranging from 0 (Big/LowBM, no overlap) to 8 (Small/HighBM, maximal overlap). The confirmatory test is a Spearman rank correlation between overlap score and $-\log_{10}(p)$ of the Granger test across 25 portfolios, with significance assessed by spatial permutation (10,000 shuffles of grid assignments; time series are never touched).

Result: $\rho_s = 0.35$, permutation $p = 0.046$ (Figure 3). Granger significance concentrates in small-cap portfolios: all five Small quintile portfolios are nominally significant ($p < 0.01$), with three surviving Bonferroni correction ($\alpha/25 = 0.002$): S/L ($p = 2.5 \times 10^{-6}$, $\Delta R^2 = 1.47\%$), S/3 ($p = 0.0014$), and S/H ($p \approx 0$, $\Delta R^2 = 1.79\%$). One Big-quintile portfolio is nominally significant ($p < 0.01$), but none survive Bonferroni. This spatial pattern is inconsistent with a uniform factor-level effect and supports the overlap channel.

Overlap fraction. At lag 15 (matching Table 4), $\Delta R^2(\text{HML} \rightarrow \text{S/H}) / \Delta R^2(\text{HML} \rightarrow \text{SMB}) = 0.392$: the Small-HighBM portfolio alone accounts for 39% of the HML→SMB incremental R^2 .

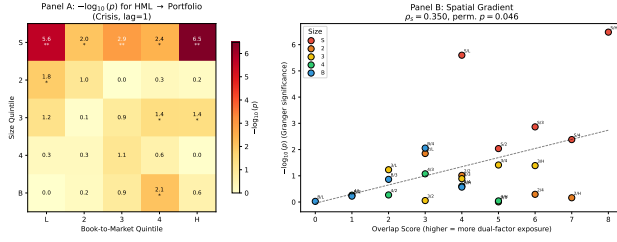


Figure 3: FF 25 Portfolio Overlap Analysis. Panel A: $-\log_{10}(p)$ for HML→Portfolio Granger tests (Crisis, lag=1). Significance concentrates in Small quintile; **Bonferroni-significant. Panel B: Spatial gradient—overlap score vs. Granger significance ($\rho_s = 0.35$, permutation $p = 0.046$).

Table 5: Early Warning Lead Time. First Detection = first day of 3+ consecutive Crisis regime assignments.

Event	First Det.	Vol. Peak	Lead
Lehman 2008	May 9, 2008	Sep 15, 2008	129 days
EU Crisis 2011	None	Aug 8, 2011	None
COVID 2020	None	Mar 23, 2020	None

4.6 EARLY WARNING PERFORMANCE

The Student- t HMM detects crisis regimes before market peaks (Table 5). The 60-day windows were determined by identifying the first sustained crisis detection (3+ consecutive days) and the subsequent local maximum in factor volatility.

5 CONCLUSION

The recomputed, leakage-safe primary fit supports a narrower conclusion than earlier drafts. HML→SMB is strongly significant in the Normal regime at lag 1 ($p = 8.75 \times 10^{-9}$; HAC $p = 9.45 \times 10^{-8}$). Elevated evidence is suggestive but non-confirmatory under the paper’s multiple-testing threshold, and Crisis evidence is null at lag 1 ($p = 0.695$). The frozen out-of-sample aggregate test is positive at conventional levels ($p = 0.041$), but event-level results remain mixed.

Does establish:

- Confirmatory Normal-regime HML→SMB predictive precedence
- Exploratory FF25 overlap evidence ($\rho_s = 0.35$, permutation $p = 0.046$)
- No evidence of tradable alpha (Sharpe = -0.07)
- Practical utility of a hybrid detector for VaR calibration (5.60% violations, CC $p = 0.336$)

Does not establish:

- Uniform significance across every regime
- Uniform event-level support across all stress episodes
- Structural causality (common drivers could explain the pattern)
- Trading profitability

Practical value: A hybrid VaR model combining HMM regime detection with a realized-volatility fallback achieves the closest coverage to the 5% target (5.60%, $p_{CC} = 0.336$), detecting 47.8% of COVID-2020 where the HMM alone detected 0%. Algorithm ?? formalizes the monitoring protocol with all parameters calibrated on training data.

For practitioners, the key insight is that factor lead-lag relationships are not merely stronger or weaker across regimes. In the recomputed primary fit, the clearest actionable signal is Normal-regime lag-1 HML→SMB. Crisis-regime monitoring is better treated as stress-sensitivity analysis than as a confirmatory directional signal. Real-time implementation still benefits from the volatility override: the frozen HMM alone classified 0% of COVID-2020 as Crisis, while the hybrid detector reached 47.8% detection with one pre-specified threshold parameter.

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